

Replication

[RE] Ramp Metering for a Distant Downstream Bottleneck Using Reinforcement Learning with Value Function Approximation

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A replication of "Ramp Metering for a Distant Downstream Bottleneck Using Reinforcement Learning with Value Function Approximation" from Zhou et al. (2020), published in the 2020 volume of the Journal of Advanced Transportation (Article ID 8813467).

Replication Summary

Scope of Replication — This project replicates and evaluates the claims made by the original study of [1], *Ramp Metering for a Distant Downstream Bottleneck Using Reinforcement Learning with Value Function Approximation*. The original article proposes a reinforcement learning (RL) based controller for ramp metering and reports improved performance when compared with conventional feedback-based ramp metering algorithms.

Methodology — As no supplementary material, source code, or numerical demand data was provided, the full experimental setup was re-implemented from scratch in Python. The simulation environment was built using a Cell Transmission Model (CTM), while the RL agent was implemented and trained using PyTorch. Missing parameters were either assumed or reconstructed from the published figures and methodological description.

Results — The no-control and reference PI-ALINEA scenarios could be reproduced closely. However, the reported performance advantage of the RL controller could not be confirmed when stronger benchmark controllers were introduced. In particular, optimised PI-ALINEA variants matched or outperformed the RL controller.

What was easy — Implementing the basic CTM dynamics and reproducing the qualitative no-control and PI-ALINEA traffic patterns was comparatively straightforward once the main simulation parameters had been identified.

What was difficult — The main difficulties were reconstructing undocumented model parameters, matching the original demand profile, implementing a stable RL training procedure, and comparing results against the original study, which reported mostly graphical outputs and no quantitative performance tables.

Communication with the original authors — No attempt was made to contact the original authors.

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The authors have declared that no competing interests exists.

Code is available at https://github.com/DerKevinRiehl/replication_ramp_metering_rl_2020_zhouetal.

1 Introduction

Reinforcement learning (RL) [2] has gained increasing attention in control and transportation research, as it enables control policies to be learned through interaction with a simulated environment [3, 4]. In intelligent transportation systems, RL has been applied to freeway control problems such as ramp metering, variable speed limits, and combined traffic management strategies [5].

Ramp metering regulates on-ramp inflow to reduce congestion and improve mainline throughput [6]. Classical feedback controllers, such as ALINEA and PI-ALINEA, use traffic measurements near the control target location to determine the metering rate [7, 8]. While these methods are well established for bottlenecks close to the ramp, distant downstream bottlenecks introduce delays between metered ramp vehicles and their effect at the bottleneck, which can reduce controller stability. This motivates control strategies that can anticipate traffic evolution between the on-ramp and the downstream bottleneck.

This work aims to reproduce the study by [1], which formulates ramp metering for a distant downstream bottleneck as a Q-learning problem with value function approximation. In the original study, the agent observes traffic densities at representative locations between the ramp and the bottleneck, together with the estimated ramp demand, and selects a discrete metering rate. An artificial neural network approximates the action-value function, allowing the method to operate in a multidimensional continuous state space without a large lookup table.

The purpose of this reproduction is to evaluate the reported performance of the proposed RL controller and to assess the reproducibility of the experimental setup. As highlighted by [9], transportation simulation studies often face reproducibility challenges due to missing source code, incomplete calibration information, and insufficiently documented modelling assumptions. Although the original article provides the main modelling concepts and controller structure, several implementation details required for exact reproduction are unspecified or only available graphically. This study therefore reconstructs the simulation and learning pipeline from the published description and documents the assumptions required to reproduce the reported results.

The remainder of this paper is structured as follows. Section 2 summarises the original study. Section 3 describes the reproduction methodology. Section 4 presents and discusses the reproduced results. Finally, Section 5 concludes this replication study.

2 Summary of Original Paper

The original study from [1] addresses the problem of ramp metering for a bottleneck located far downstream of the metered ramp. Conventional linear feedback-type controllers, such as ALINEA [7] or PI-ALINEA [8], suffer from stability issues in this setting due to the significant time-delay between the ramp and the distant bottleneck: by the time metered vehicles reach the bottleneck, its traffic state may have changed substantially from when it was sampled to compute the metering rate. Rather than compensating for this delay through predictors of traffic flow evolution, the authors propose formulating the problem as a Q-learning problem in which an agent learns a non-linear feedback policy directly from interaction with a traffic simulation environment.

The state vector consists of four continuous variables: traffic densities measured at three representative locations along the stretch between the ramp and the downstream bottleneck (ρ_1, ρ_2, ρ_3), and an estimated ramp demand D_{ramp} . The agent selects a discrete metering rate at each time step, with the action space bounded above by the estimated ramp demand to avoid unrealistic control actions. The reward function penalises deviations of the downstream bottleneck density ρ_3 from a desired critical density value. To handle the computational burden of the four-dimensional continuous state space,

the Q-value function is approximated by an artificial neural network (ANN) rather than a lookup table, reducing the number of parameters to learn from an estimated 25.6 million Q-values to approximately 67,000. State variables are first encoded via a tile coding scheme into a 140-dimensional binary feature vector before being passed to the ANN. Parameters are updated online at each time step using an incremental back-propagation algorithm [10].

The approach is validated on a synthetic freeway section implemented in the Cell Transmission Model (CTM), consisting of a three-lane mainline with a lane-drop to two lanes located 3500 m downstream of the metered ramp. Learning is reported to have converged after approximately 0.7 million episodic iterations. The proposed method is compared against an uncontrolled baseline and a conventional PI controller (PI-ALINEA [8]). Results show that the learned policy successfully maintains the bottleneck density close to the desired value with negligible oscillations, whereas the PI controller exhibits some instability at this distance. A robustness analysis under additive white noise on traffic demands shows satisfactory performance up to a standard deviation of 200 veh/h.

3 Implementation

3.1 Building the Simulation Environment

The simulation environment was implemented using the first-order macroscopic Cell Transmission Model (CTM) as introduced by [11]. The CTM discretises a freeway section into homogeneous cells of equal length Δx and evolves traffic density forward in time according to:

$$\rho_i(t+1) = \rho_i(t) + \frac{\Delta t}{\Delta x} [y_i(t) - y_{i+1}(t)] \quad (1)$$

where $\rho_i(t)$ denotes the traffic density in cell i at time t , Δt the time interval between the steps $[t; t+1]$, and $y_i(t)$ denotes the flow entering cell i from its upstream neighbour. The inter-cell flow is computed from separate sending and receiving functions:

$$y_i(t) = \min\{S_{i-1}(t), R_i(t)\} \quad (2)$$

where:

$$S_i(t) = \min\{v_f \rho_i(t), Q_i\} \quad (3)$$

$$R_i(t) = \min\left\{Q_i, w\left(\rho_i^{\text{jam}} - \rho_i(t)\right)\right\} \quad (4)$$

Here v_f denotes the free-flow speed, w the backward congestion-wave speed, Q_i the capacity of cell i , and ρ_i^{jam} its jam density. The triangular fundamental diagram assumed throughout implies:

$$w = \frac{Q_{\text{lane}}}{\rho_{\text{lane}}^{\text{jam}} - \rho_{\text{lane}}^{\text{cr}}} \quad (5)$$

An important restriction lies in the spatial and temporal discretisation, which must be selected to satisfy the Courant–Friedrichs–Lewy stability condition (CFL) for explicit finite-difference schemes [12], which in the CTM context requires that vehicles cannot travel more than one cell during a single time step:

$$\Delta x = v_f \Delta t \quad (6)$$

Model parameters. The original study does not report the values of Δx or Δt used in its simulations. To recover a consistent parameterisation, the model was implemented in a modular fashion so that both quantities could be adjusted. A systematic exploration of feasible pairs satisfying the CFL condition was conducted, and results were compared against the no-control density contour reported by the authors. A cell length of $\Delta x = 500$ m with a corresponding time step of $\Delta t = 15$ s was found to reproduce the reported congestion patterns satisfactorily.

The remaining fundamental diagram parameters were set as stated in the original work: free-flow speed $v_f = 120$ km/h, critical density $\rho_{\text{lane}}^{\text{cr}} = 20$ veh/(km·lane), and jam density $\rho_{\text{lane}}^{\text{jam}} = 100$ veh/(km·lane), yielding a per-lane capacity of $Q_{\text{lane}} = 2400$ veh/h. Cell capacities and jam densities were scaled linearly with the local number of lanes. The network layout follows Figure 3 of the original study: a 6 km freeway section with a metered on-ramp at 2 km and a lane-drop from three to two lanes at 5.5 km downstream. The simulation horizon for each episode was set to 6000 s.

Boundary conditions. The upstream boundary was implemented as a demand reservoir rather than as a physical road cell. The inflow into the first road cell is therefore not constrained by an upstream sending capacity, but by the available mainline demand, any accumulated upstream queue N_2 , and the receiving capacity of the first cell:

$$y_1(t) = \min \left\{ d_{\text{main}}(t) + \frac{N_2(t)}{\Delta t}, R_1(t) \right\} \quad (7)$$

Any mainline flow that cannot enter the network accumulates in the upstream queue:

$$N_2(t+1) = \max \{ 0, N_2(t) + (d_{\text{main}}(t) - y_1(t)) \Delta t \} \quad (8)$$

At the downstream end of the network, free outflow was assumed. The outflow from the final cell is determined solely by its sending function:

$$y_n(t) = S_n(t) \quad (9)$$

implying that congestion originating outside the simulated network cannot propagate upstream into the modelled section.

On-ramp entry cell. The original study does not unambiguously specify into which cell the on-ramp flow is injected. Based on Figure 3 of the original work, the ramp is located at approximately 2 km, which falls on the boundary between two cells when $\Delta x = 500$ m. This ambiguity was resolved by careful inspection of the spatio-temporal density contours reported in the original study. Under the PI-ALINEA scenario, five characteristic congestion waves are visible in the authors' results, propagating downstream and starting at the 1500 m mark (see Figure 6). This pattern is only consistent with ramp flow being injected into cell 4, corresponding to the segment 1500–2000 m, and was therefore adopted throughout all experiments. The figure below (1) pictures the resulting experimental layout used.

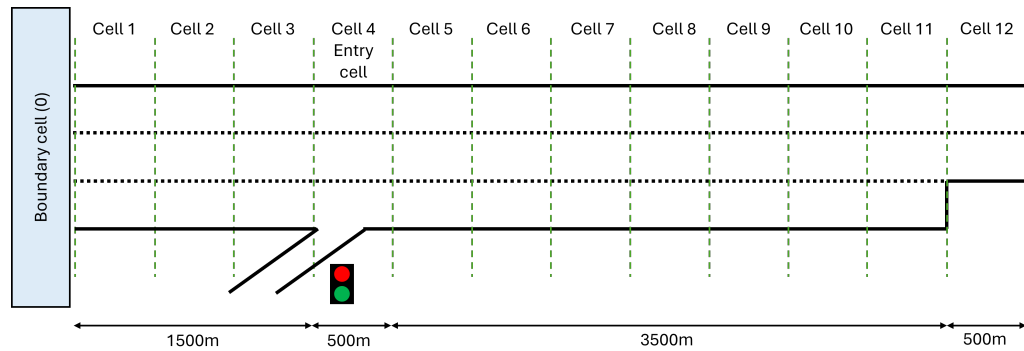


Figure 1. Layout of the freeway section used for the reproduction

Demand profile. The traffic demand profiles for the mainline and the on-ramp were not provided numerically in the original study; only a figure of limited resolution was available. The profiles were reconstructed through manual inspection of that figure and iterative adjustment until the simulated no-control and PI-ALINEA control cell density diagram (see Figure 3 and 7) and the PI-ALINEA ramp-metering rate (see Figure 8) satisfactorily matched the published result.

The reconstructed profiles (see Figure 2) follow a trapezoidal shape: a linear ramp-up of 600 s, a plateau of 1800 s, and a symmetric ramp-down of 600 s, superimposed on constant baseline demands of 1000 veh/h for the mainline and 500 veh/h for the on-ramp. Peak demand increments of 3000 veh/h and 720 veh/h were applied to the mainline and on-ramp respectively, yielding maximum demands of 4000 veh/h and 1220 veh/h.

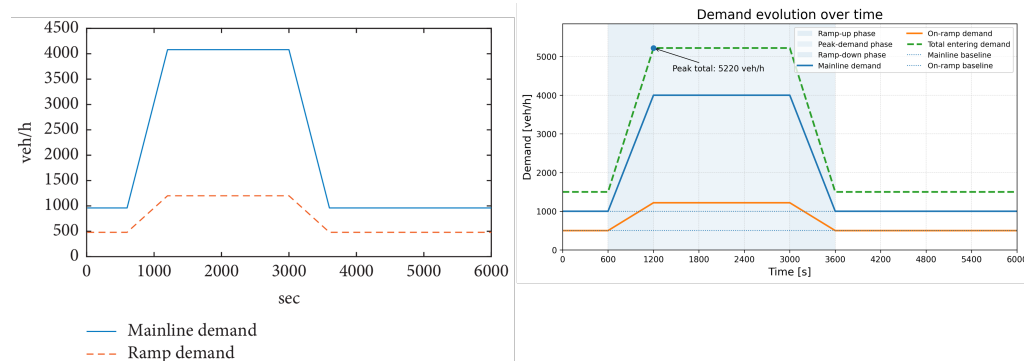


Figure 2. Comparison of demand evolution in original study (left) and reproduction (right)

To facilitate exact reproduction, future studies are encouraged to report demand profiles numerically alongside the simulation parameters Δx and Δt , and to document explicitly the cell at which boundary flows are injected.

3.2 Implementation of the benchmark controllers

No-Control Scenario – As a baseline for comparison, an uncontrolled scenario was implemented in which no ramp metering strategy was applied. In the absence of control actions, all on-ramp demand was released whenever sufficient downstream receiving capacity was available. If the combined mainline and ramp demand exceeded the receiving capacity of the downstream cell, the available capacity was distributed proportionally between the two streams.

The resulting traffic evolution serves as a reference case against which the effectiveness of the controllers can be evaluated.

3.3 Benchmark Controllers: ALINEA and PI-ALINEA

Two classical feedback-type ramp metering controllers were implemented as benchmarks: ALINEA [7] and its proportional–integral extension PI-ALINEA [8]. PI-ALINEA is the sole benchmark considered in the original study; ALINEA was additionally implemented in this replication as a second more widely used baseline. Both controllers share the same underlying control philosophy: regulate the on-ramp inflow so as to maintain the traffic density at a designated control location close to a desired setpoint, thereby preventing congestion formation at the downstream bottleneck while maximising throughput.

ALINEA. ALINEA is a local proportional feedback controller that adjusts the metering rate based solely on the deviation of the measured density from the desired value. The control law reads:

$$r(k) = r(k-1) + K_R[\hat{\rho} - \rho(k)] \quad (10)$$

where $r(k)$ denotes the metering rate applied during control interval k , $\rho(k)$ is the measured density at the control location, $\hat{\rho}$ is the desired density, and K_R is the integral gain. The controller increases the metering rate when density falls below the set-point and reduces it when density exceeds it, with no memory of past density errors beyond the previous metering rate $r(k-1)$.

PI-ALINEA. PI-ALINEA extends ALINEA by adding a proportional term that reacts to the rate of change of density, improving stability in the presence of time delays such as those arising when the controlled bottleneck is located far downstream of the metered ramp [8]. The control law is:

$$r(k) = r(k-1) + K_R[\hat{\rho} - \rho(k)] + K_P[\rho(k-1) - \rho(k)] \quad (11)$$

where K_P is the proportional gain and all other quantities are defined as above. When $K_P = 0$ the expression reduces exactly to the ALINEA law, making the two controllers structurally identical up to this single additional term.

Implementation details. For both controllers, the control cell was set to cell 11, the last three-lane cell immediately upstream of the lane-drop bottleneck (see Figure 1), consistent with the formulation of the original study. The desired density was set to $\hat{\rho} = 13.33 \text{ veh}/(\text{km}\cdot\text{lane})$, corresponding to the critical density scaled by the lane-drop ratio $(\lambda_2/\lambda_1) \rho_{\text{lane}}^{\text{cr}} = (2/3) \times 20$, as reported by the authors. Furthermore, in both cases, the metering rate was clipped to the admissible range $[r_{\min}, r_{\max}] = [200, 1200] \text{ veh/h}$ at every control step as stated in the original article to fit Full Traffic Cycle (FTC) restrictions according to [13].

The original publication does not provide details of any controller calibration procedure. Consulting the PI-ALINEA reference [8] directly, the recommended gains for a distant bottleneck configuration are $K_R = 4$ and $K_P = 100$, which were adopted without modification for PI-ALINEA.

Optimizing the Benchmark Controllers – In addition to reproducing the benchmark controllers with the parameter values used in the reference studies, a systematic grid search was conducted to assess whether improved controller parameters could be identified. Two optimisation criteria were considered: one evaluating mainline performance only, and one additionally penalizing on-ramp queueing delay. The distinction reflects a fundamental trade-off in ramp metering — aggressive metering improves mainline flow but at the cost of accumulating ramp queues — and ensures that the benchmark controllers are assessed from both a mainline and a system-wide perspective.

Performance indicators. Both criteria are derived from vehicle-kilometres travelled (VKT) and vehicle-hours travelled (VHT). For a given simulation run, VKT was computed as:

$$\text{VKT} = \sum_{k=1}^{N_t} \sum_{i=1}^N q_i(k) \Delta x \Delta t \quad (12)$$

where $q_i(k)$ is the outflow of cell i at time step k , Δx is the cell length, and Δt is the simulation time step expressed in hours. The mainline VHT was computed from the accumulated number of vehicles present in the freeway cells:

$$\text{VHT} = \sum_{k=1}^{N_t} \sum_{i=1}^N \rho_i(k) \Delta x \Delta t \quad (13)$$

The first optimisation criterion, targeting mainline performance only, is the average network speed:

$$\bar{v} = \frac{\text{VKT}}{\text{VHT}} \quad (14)$$

which serves as a proxy for minimizing mainline total time spent (assuming there is no queue at the end of the simulation). Since this criterion does not account for vehicles delayed in the on-ramp queue, a queue-aware variant was introduced by augmenting VHT with the accumulated ramp queue time:

$$\text{VHT}_{\text{queue}} = \text{VHT} + \sum_{k=1}^{N_t} N_1(k) \Delta t \quad (15)$$

where $N_1(k)$ denotes the number of vehicles waiting in the on-ramp queue at time step k . The queue-aware average speed

$$\bar{v}_{\text{queue}} = \frac{\text{VKT}}{\text{VHT}_{\text{queue}}} \quad (16)$$

constitutes the second optimisation criterion, providing a more system-oriented measure of performance. When no residual ramp queue remains at the end of the simulation, $\bar{v}_{\text{queue}}$ closely approximates total system performance. If a queue persists beyond the simulation horizon, however, the true total time spent is still underestimated as the future delay of unserved ramp vehicles is not captured.

Grid search. For ALINEA, the regulator gain was varied over the interval:

$$K_R \in \{1, 2, \dots, 110\}. \quad (17)$$

For PI-ALINEA, both gains were varied over the same range:

$$K_R \in \{1, 2, \dots, 110\}, \quad K_P \in \{1, 2, \dots, 110\}. \quad (18)$$

Each parameter combination was evaluated over the full simulation horizon of 6000 s. The best-performing ALINEA and PI-ALINEA parameters were identified separately under both optimisation criteria. These retuned controllers were then compared with the original parameter choices and with the RL-based controller in the subsequent evaluation.

3.4 Building the RL-Agent

The RL agent was implemented following Algorithm 1 of the original article and the architectural specifications of Section 3.4. The agent observes a four-dimensional continuous state vector at each time step, consisting of per-lane traffic densities sampled at three representative locations along the freeway stretch— ρ_1 at the upstream end of the stretch (cell 5, immediately downstream of the metered ramp), ρ_2 at the midpoint of the stretch (cell 8), and ρ_3 at the control target cell (cell 11, immediately upstream of the lane-drop)—alongside the estimated ramp demand D_{ramp} computed from the current on-ramp queue length and the previous time step's actual ramp inflow, following equation 1 of the original work. For a graphic representation of the cells locations, see Figure 1. Note that the exact cell numbers are not explicitly stated in the source article, but could be easily identified with the help of the original figure 1 once the cell dimensions were identified.

Prior to being fed into the neural network, the continuous state vector is encoded into a 140-dimensional binary feature vector using a tile coding scheme [3]; each of the three density variables is mapped to one of 40 equal-width bins covering the range $[0, \rho_{\text{jam, lane}}] = [0, 100]$ veh/km/lane, and D_{ramp} is mapped to one of 20 bins, the first 19 covering $[0, a_{\text{max}}] = [0, 1200]$ veh/h uniformly and the last bin capturing any value above a_{max} , the highest possible action. This follows the encoding described in the study. However, as the on-ramp demand never falls below 500 veh/h during training, 37% of the encoding space of the ramp demand ends up never getting used during training. A simple, more optimised alternative would be to cover the interval $[0; 500]$ veh/h with 1 bin, $[500, 1200]$ veh/h with 18 bins, and 1 final bin for values above a_{max} .

The approximate value function is a fully connected artificial neural network (ANN) with a 140-dimensional input layer, a single hidden layer of 420 neurons, and an output layer of 11 neurons corresponding to the 11 discrete admissible metering rates $\{200, 300, \dots, 1200\}$ veh/h, to again accommodate FTC recommendations [13]. The hidden layer uses a sigmoid activation function, as explicitly specified in the original paper. The output layer is linear, so Q-values can take any real value, positive or negative, unbounded.

A critical implementation challenge arose from the combination of sigmoid activation, online stochastic gradient descent (SDG, i.e., incremental back-propagation), bootstrapped Q-learning targets, and the unbounded linear output layer for the Q-values. Sigmoid activations can saturate for large input magnitudes, causing their derivatives to become very small and thereby degrading gradient propagation [14]. Since the original article only states that the ANN parameters were initialised with small random values, without specifying a reproducible initialisation scheme, Xavier–Glorot uniform initialisation was used for both linear layers [14]. In addition, gradient clipping with a maximum norm of 1.0 was applied at each update step to limit occasional exploding gradients [15]. The network is trained using online SGD—one gradient step per time step—with a two-phase learning rate schedule: $\alpha = 0.05$ for the first 100,000 episodes and $\alpha = 0.01$ thereafter, following the paper's specification. The discount factor is $\gamma = 0.95$, as stated in the paper. The reward signal at each step is:

$$r = k \cdot |\rho_3 - \rho_{\text{des}}| \quad (19)$$

where k is a user-defined negative scaling constant. The paper does not specify the value of k ; in this implementation, $k = -1.0$ was used. Action selection follows an ϵ -greedy policy with ϵ decaying over the course of training. At each time step, the admissible action set is constrained from above by the estimated ramp demand D_{ramp} , preventing the agent from selecting a metering rate that exceeds the available ramp flow.

Training was conducted for 700,000 episodes, each corresponding to one full simulation of the 6000 s demand scenario using the Cell Transmission Model as the environment. The original article reports that learning converged after approximately 0.7 million episodes, but does not specify the precise convergence criterion or the decay schedule used for the ϵ -greedy exploration policy. Therefore, training was run for the reported

700,000 episodes in all experiments.

Since the exploration decay was not specified, an exponential decay factor of 0.99999 was adopted for ϵ , starting from $\epsilon_0 = 1.0$. With this setting, the exploration rate reaches approximately 5% after about 300,000 episodes. Two exploration variants were tested: one in which ϵ was allowed to continue decaying without a lower bound, and one in which a minimum exploration rate of $\epsilon_{\min} = 0.05$ was retained. The latter follows common reinforcement learning practice, where a non-zero exploration floor is often used to avoid premature convergence to a deterministic policy [16].

Network checkpoints were saved every 10,000 episodes to allow recovery from interruptions and to enable retrospective evaluation of policy quality at different stages of training. Software and computational environment specifications can be seen in the appendix A.

4 Results and Discussion

The original study provides graphical results for three scenarios only: the no-control case, the non-optimised PI-ALINEA controller, and the RL agent, all evaluated under the base demand profile. A robustness analysis is additionally presented for the RL agent under five levels of additive white noise on the traffic demand (standard deviation of 50, 100, 150, 200, and 250 veh/h). No quantitative performance indicators of any kind are reported in the original work.

In this replication, all controllers were evaluated under the base demand scenario as well as under each of the five noise levels, enabling a systematic cross-controller comparison that was absent from the original study. The quantitative performance indicators recorded for each scenario are summarised in Table 1.

Table 1. Performance indicators recorded for each controller and demand scenario.

Indicator	Description
VKT	Vehicle-kilometres travelled
VHT	Vehicle-hours travelled (mainline only)
VHT _{queue}	Vehicle-hours travelled including ramp queue delay
$\bar{v}$	Average speed, mainline only
$\bar{v}_{\text{queue}}$	Average speed including ramp queue delay
Episode reward	Cumulative reward following Eq. 19

A complete overview of all recorded performance values across controllers and demand scenarios is provided in Table 4. Table 2 reports the percentage gaps between controllers, which is computed relative to the best controller for each metric and demand scenario; the calculation procedure is given in the Appendix (B). These tables constitute the main quantitative contribution of this replication and are discussed in detail in the following subsections.

4.1 Reproducing Baseline Results

No-Control Scenario – Figure 3 shows the reproduced control-cell density alongside the corresponding result reported in the original study. Figures 4 and 5 present the reproduced space-time density contours in smoothed and non-smoothed form respectively. The comparisons in Figures 3 and 4 were used to fine-tune the traffic demand profile and CTM parameters, and show a strong overall resemblance to the original results, confirming that the simulation environment has been reconstructed faithfully. Figure 5 provides a direct view of the underlying simulation output without post-processing, which is absent from the original study. Raw density contours are preferred for reproducibil-

ity purposes, as smoothing operations are difficult to replicate exactly and may obscure differences between results.

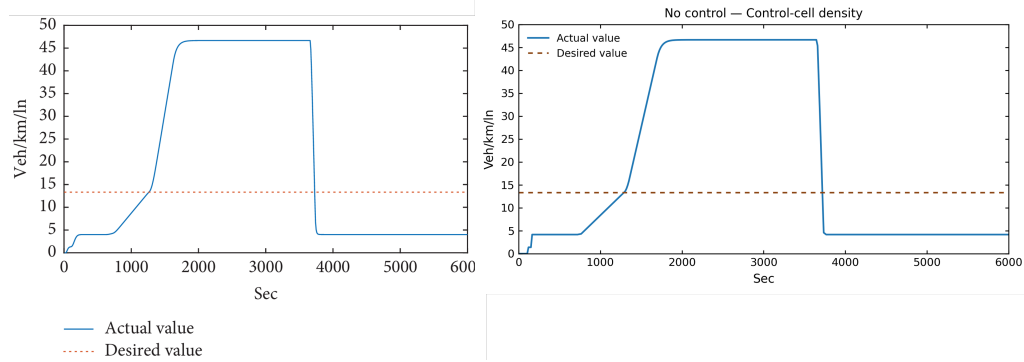


Figure 3. No-control scenario: control-cell density. Original study (left) and reproduced result (right).

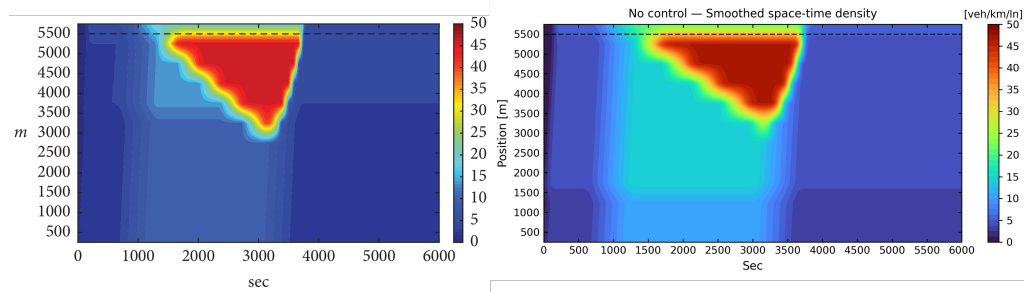


Figure 4. No-control scenario: smoothed space-time traffic density. Original study (left) and reproduced result (right).

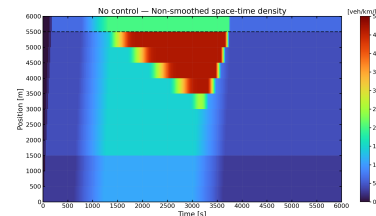


Figure 5. No-control scenario: non-smoothed space-time traffic density of the reproduced result.

PI-ALINEA Scenario – Figures 6, 7, and 8 compare the reproduced PI-ALINEA results against those reported in the original study for the space-time density contour, the control-cell density trajectory, and the ramp metering rate respectively. The parameters used are those adopted from the reference study: $K_R = 4$ and $K_P = 100$ [8].

All three figures show close agreement with the original results. In particular, the oscillatory pattern of the metering rate and the characteristic suppression of congestion at the bottleneck are well reproduced. Minor discrepancies in the smoothed density contours are attributed to differences in the post-processing procedure, which was not described in the original study. These results confirm that the PI-ALINEA controller has been correctly implemented and that the simulation environment produces consistent outputs under controlled conditions.

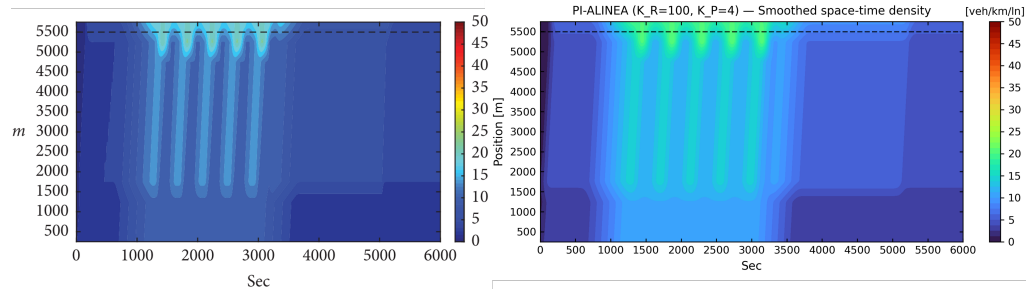


Figure 6. Base PI-ALINEA scenario: smoothed space-time traffic density. Original study (left) and reproduced result (right).

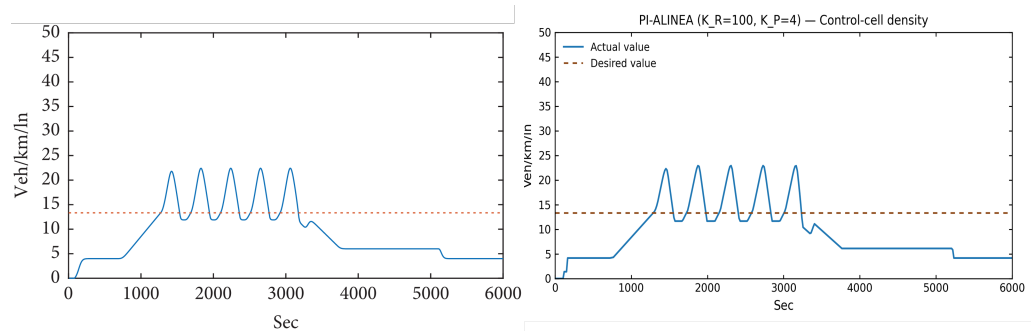


Figure 7. Base PI-ALINEA scenario: control-cell density. Original study (left) and reproduced result (right).

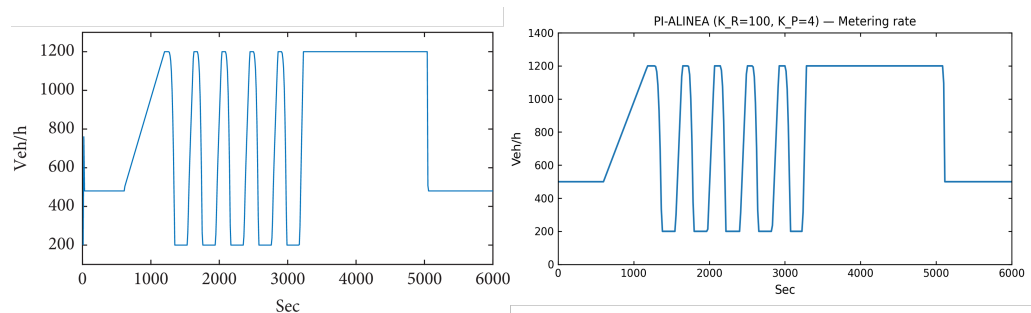


Figure 8. Base PI-ALINEA scenario: on-ramp metering rate. Original study (left) and reproduced result (right).

4.2 Optimizing Benchmark Controllers

The grid search identified the following optimal parameter configurations. For PI-ALINEA, the best parameters under the mainline criterion $\bar{v}$ were $K_R = 2$ and $K_P = 93$, yielding $\bar{v} = 119.85$ km/h. Optimizing instead for the queue-aware criterion $\bar{v}_{\text{queue}}$ produced $K_R = 2$ and $K_P = 56$, with $\bar{v}_{\text{queue}} = 78.90$ km/h. For ALINEA, the optimal gain under $\bar{v}$ was $K_R = 23$ ($\bar{v} = 119.37$ km/h), and under $\bar{v}_{\text{queue}}$ was $K_R = 21$ ($\bar{v}_{\text{queue}} = 72.70$ km/h). The similarity of the optimal K_R values between the two criteria for both controllers suggests that the integral gain plays a secondary role relative to K_P in PI-ALINEA, and that performance is relatively robust to small variations in K_R for ALINEA.

The optimised controllers can first be evaluated graphically, following the methodology of the original study, using Figures 9, 10, and 11, which show the space-time density contours, control-cell density trajectories, and ramp metering rates under the base demand scenario.

Overall, the graphical comparison suggests that all optimised benchmark controllers improve upon the non-optimised PI-ALINEA baseline. The improvement is most visible in the metering-rate trajectories shown in Figure 11, where the optimised controllers produce substantially more stable ramp inflows than the oscillatory metering pattern obtained with the reference PI-ALINEA parameters in Figure 8. This effect is particularly clear for the optimised PI-ALINEA configurations, whose metering rates vary more smoothly while still preventing severe congestion at the downstream bottleneck.

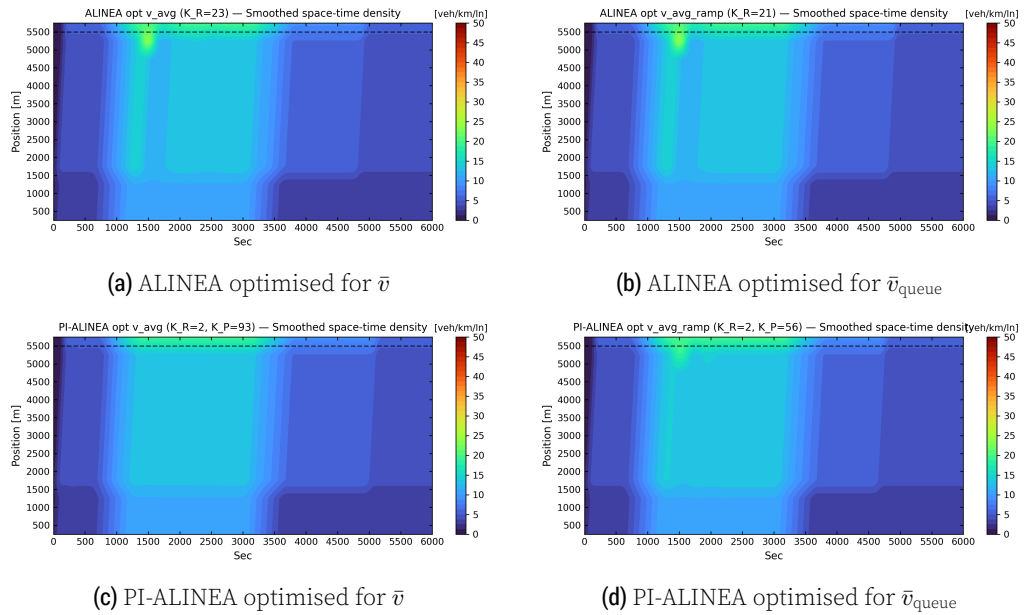


Figure 9. Optimised benchmark controllers: space-time density contours under base demand.

The control-cell density trajectories in Figure 10 further indicate that the optimised PI-ALINEA controllers are more effective at maintaining the density close to the desired value. In particular, the PI-ALINEA controller optimised for $\bar{v}$ appears to keep the control-cell density closest to ρ_{des} , suggesting that optimizing this controller to maximise the mainline average speed also partly coincides with maximizing the density-based reward used by the RL agent. This is consistent with the definition of the reward function, which directly penalises deviations of the control-cell density from the desired density.

The space-time density contours in Figure 9 support the same interpretation. The optimised controllers limit the formation of high-density regions near the lane-drop and produce smoother traffic patterns than the reference PI-ALINEA configuration. The

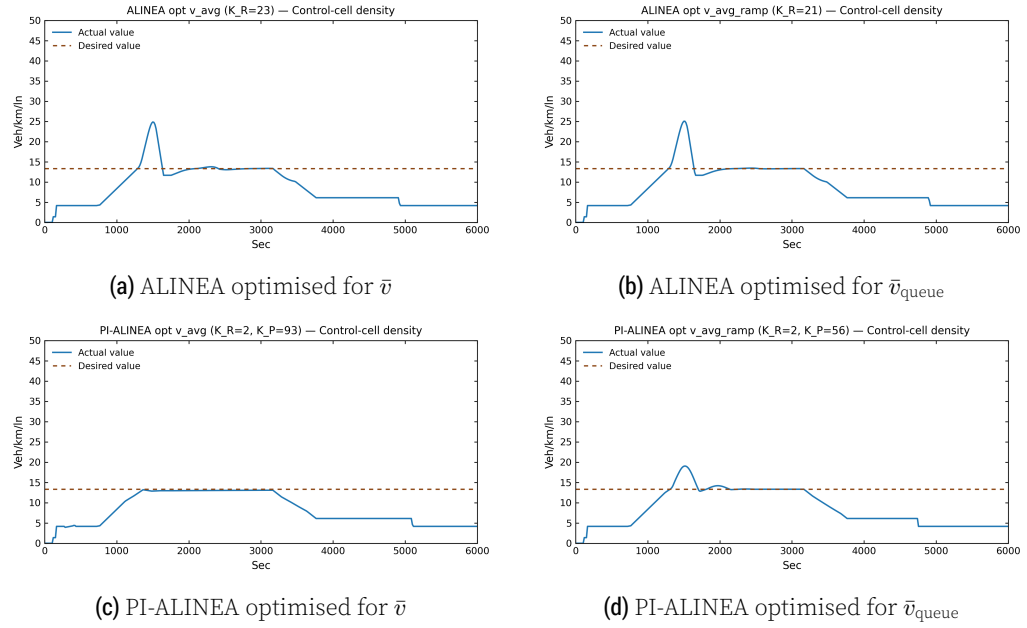


Figure 10. Optimised benchmark controllers: control-cell density under base demand.

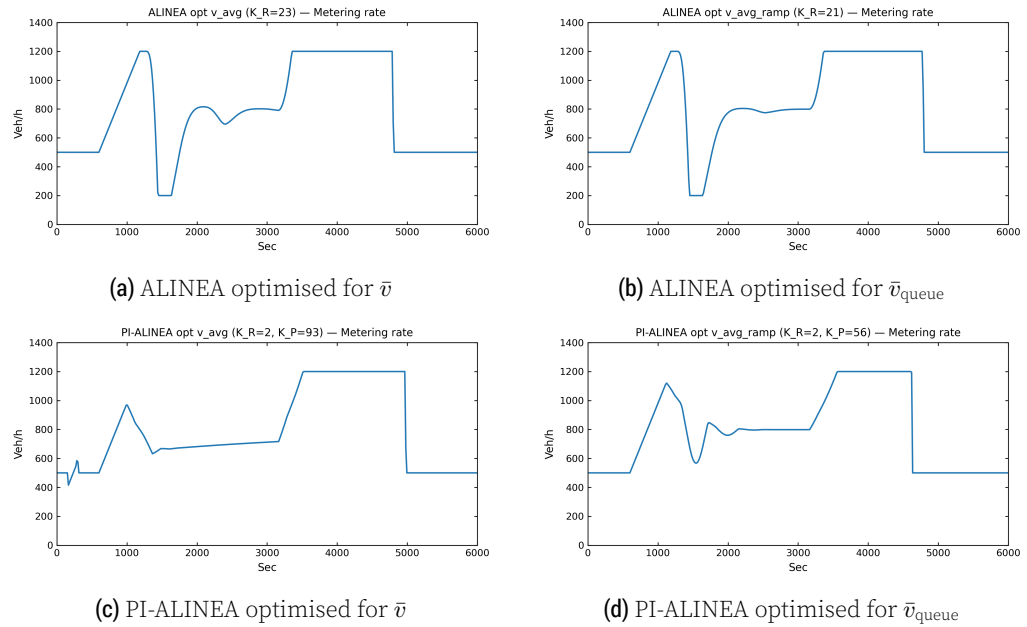


Figure 11. Optimised benchmark controllers: on-ramp entry rate under base demand.

queue-aware optimisation criterion $\bar{v}_{\text{queue}}$ leads to slightly more conservative control settings, as expected, because the objective also penalises ramp waiting time. Quantitative performance of all controllers across all demand scenarios is reported in Table 4.

4.3 RL Agent

Training Behaviour – A total of six independent RL training campaigns were conducted. Four campaigns used an ϵ -greedy exploration policy with a lower exploration bound of $\epsilon_{\min} = 0.05$, while two campaigns allowed ϵ to decay without a lower bound. Since the original article does not specify the ϵ -decay schedule or the convergence criterion used during training, all campaigns were run up to the reported training length of 700,000 episodes.

Figure 12 shows representative training curves for one capped and one uncapped exploration campaign. The remaining campaigns showed similar qualitative behaviour. In general, the training process did not converge towards a stable final policy. Instead, policy quality varied across checkpoints, with the final networks at 700,000 episodes not necessarily being the best-performing policy.

Both the learning curve and the 1000-episode rolling reward standard deviation show that the capped exploration policy reaches a plateau after approximately 200,000–300,000 episodes. Beyond this point, the rolling mean remains within a relatively stable reward interval, which is consistent with the exploration probability reaching its lower bound of $\epsilon_{\min} = 0.05$. Continued exploration therefore prevents the policy from becoming fully deterministic and maintains persistent variability in the episode rewards.

In the uncapped case, the reward variance decreases more strongly as ϵ approaches zero, indicating a transition towards an increasingly deterministic policy. However, the absence of continued exploration also makes the training process more sensitive to occasional poor updates. Once the policy has become nearly deterministic, such updates can lead to a degradation of the learned behaviour from which recovery is difficult. This behaviour is visible in the later part of the uncapped learning curve (see Figure 12).

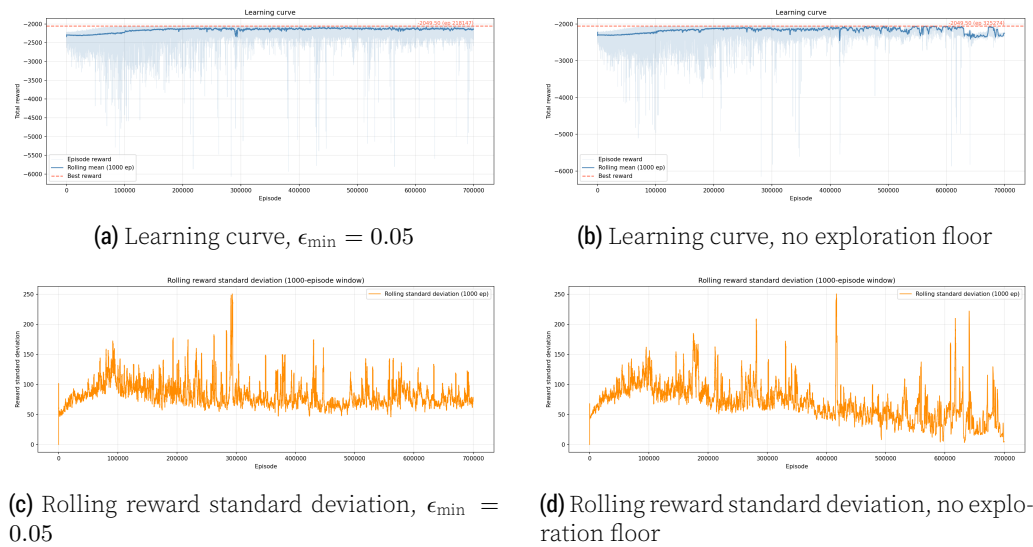


Figure 12. Representative RL training behaviour for exploration with and without a lower exploration bound. The top row shows the episode reward and rolling mean, while the bottom row shows the rolling standard deviation of the training reward.

While no convergence towards an optimal solution could clearly be observed with either exploration scheme, it was noted that the six training runs all reached the same best observed training episode reward of -2049.5 (see Figure 12). This suggests that the

corresponding episode may represent a near-optimal metering trajectory for the reconstructed demand scenario. However, since this reward was obtained during ϵ -greedy training, it does not imply that the ANN learned a deterministic policy capable of reproducing this trajectory.

Checkpoint Selection – The observations above motivated an additional checkpoint-selection procedure. Since the final checkpoint was not necessarily the best-performing policy, and since the best training episode may partly depend on exploratory actions, both exploration variants were rerun with deterministic greedy evaluation performed after every episode. In this evaluation step, the ANN selected only the action with the highest predicted value, without ϵ -greedy exploration. This procedure was not specified in the original study.

In these additional runs, both exploration variants reached the same best greedy episode reward of -2062.83 . The corresponding ANN checkpoint was selected from one of these equivalent best-performing policies for the final comparison. This reward remained below the best exploratory training reward of -2049.5 , indicating that the best metering trajectory encountered during training was not recovered by the ANN as a deterministic greedy policy.

Performance of the selected RL Policy – The most performant RL policy was then evaluated under the same base demand scenario used for the benchmark controllers. Figures 13, 14 and 15 show the resulting control-cell density, on-ramp metering rate, and space-time density contour. These figures are compared against the corresponding results reported in the original study.

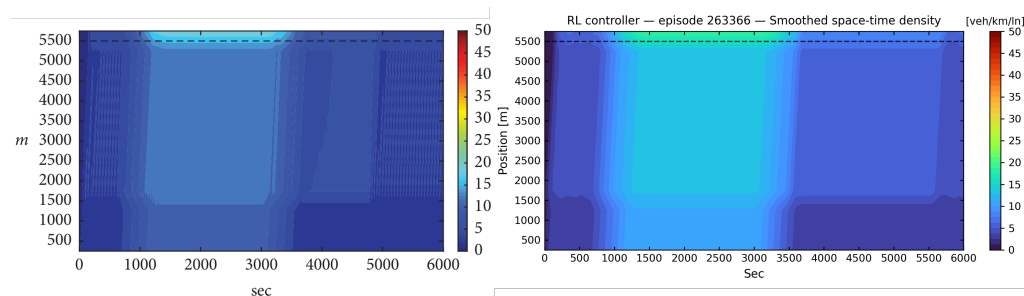


Figure 13. RL-controller scenario: Space-time density contours. Original study (left) and reproduced result (right).

Overall, the space-time density contours in Figure 13 indicate that both policies produce similar, stable and smooth traffic patterns, with densities remaining below the critical density throughout most of the simulated freeway section. However, the control-cell density trajectories in Figure 14 reveal a remaining difference between the original and reproduced controllers. The reproduced policy appears to release excess traffic more slowly, as the control-cell density remains above 5 veh/km/lane well after 5000 s.

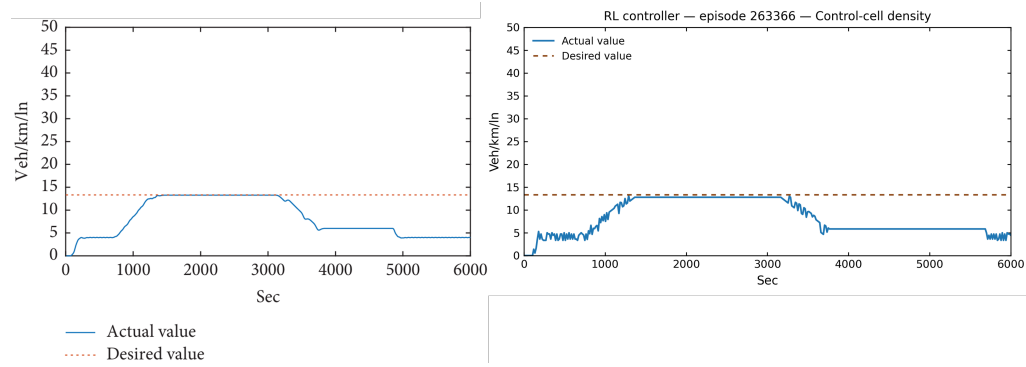


Figure 14. RL-controller scenario: control cell density. Original study (left) and reproduced result (right)

This interpretation is supported by the metering-rate comparison in Figure 15. Both policies exhibit similar stable metering intervals, but the reproduced policy is less aggressive during these intervals, allowing a lower ramp inflow than the policy reported in the original study. This lower release rate explains why the downstream density decreases more slowly and remains elevated for a longer period towards the end of the simulation.

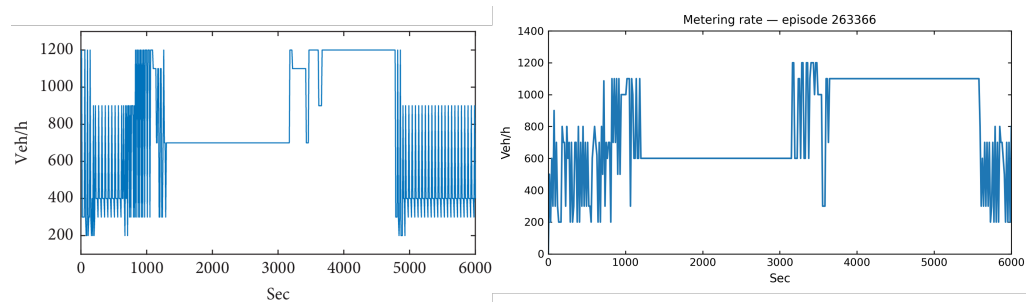


Figure 15. RL-controller scenario: on-ramp metering rate. Original study (left) and reproduced result (right)

4.4 Controller performance evaluation

Table 4 summarises the performance indicators for all controllers across the base demand scenario and five noise levels. Table 2 reports the percentage gap of each controller relative to the best-performing controller for each indicator and demand scenario. A value of 0.0% therefore indicates the best result within a given noise level and performance indicator, while larger values indicate worse performance relative to that best value. This representation makes the comparison easier to interpret than the absolute values alone, because it shows not only which controller performs best, but also how far the remaining controllers are from the best observed result. This data enables a quantitative comparison of the different controllers which is absent from the original study.

Across all controllers and demand scenarios, VKT remains effectively identical (see Table 2). This indicates that all controllers are able to serve the same total traffic demand over the simulation horizon and that no residual on-ramp queue remains at the end of the episode. Consequently, differences in performance are not caused by differences in the number of vehicles discharged from the system, but by how efficiently these vehicles are processed over time, as reflected by VHT, VHT_{queue} , $\bar{v}$, and $\bar{v}_{queue}$. The small discrepancies observed in the RL controller's VKT across noise scenarios (see Table 4) are attributable to its finite discrete action space, which prevents exact matching of the ramp demand at every time step and leads to the formation of negligible residual queues.

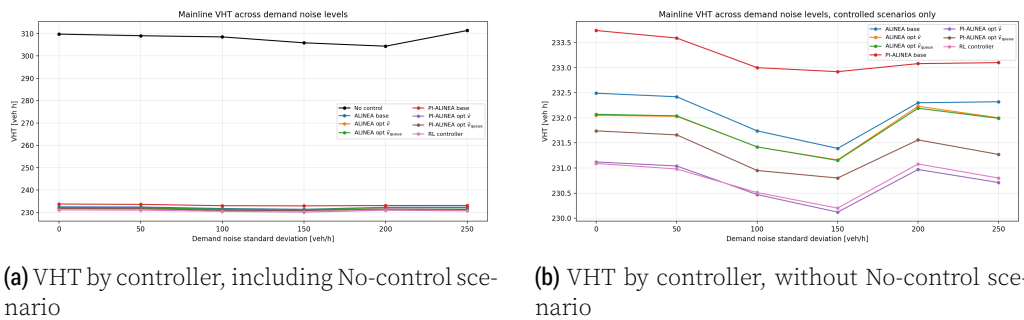


Figure 16. Evolution of VHT by controller over increasing standard deviation of the demand

Mainline travel efficiency, measured by VHT and $\bar{v}$ without ramp queue consideration, improves substantially under all controlled scenarios relative to the no-control baseline (see Figure 16). The controlled scenarios achieve an approximately 25% reduction in VHT and a 33% increase in average speed (see Table 2). Differences between controlled strategies are, however, minor, with all metering controllers achieving approximately $\bar{v} \approx 119$ km/h, regardless of the selected gains or controller variant. This suggests that once congestion at the bottleneck is prevented, the specific control law has only limited influence on mainline efficiency. Non-the-less, the original PI-ALINEA controller stands out as the worst of all controllers. However, the PI-ALINEA optimised for $\bar{v}$ either beats or matches the RL-controller across all demand scenarios.

Figure 17 shows VHT_{queue} across all demand noise levels, both with and without the no-control scenario. Compared with the mainline-only VHT results, the queue-inclusive indicator varies much more strongly across controllers and noise levels. While the controlled strategies differ only marginally in VHT, small improvements in mainline travel efficiency can correspond to substantially larger changes in VHT_{queue} . This indicates that optimizing directly for mainline VHT or $\bar{v}$ alone can worsen total system performance if the resulting policy achieves mainline stability by holding too many vehicles on the ramp.

The no-control case performs best when queue dwelling time is included, because it does not meter the ramp and therefore does not create an on-ramp queue. This raises

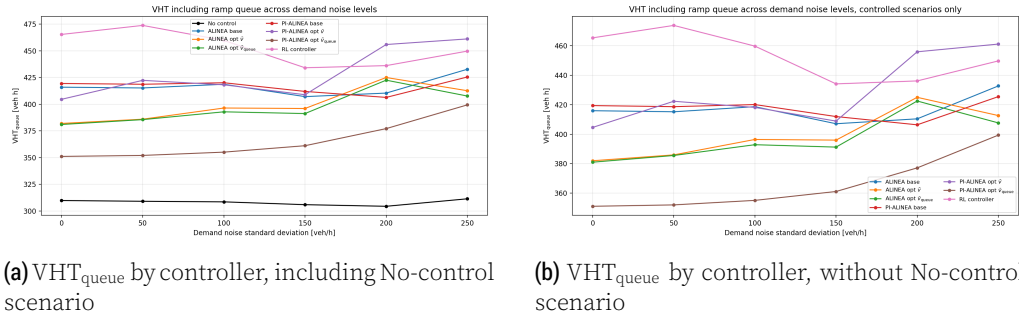


Figure 17. Evolution of VHT_{queue} by controller over increasing standard deviation of the demand

an important practical point: if the objective is defined only as total system travel time in this reconstructed distant-bottleneck scenario, ramp metering is not automatically favoured. However, this result must be interpreted together with the control objective. Ramp metering is intended not only to minimise travel time, but also to regulate mainline density, prevent bottleneck breakdown, and stabilise traffic near the lane drop. The no-control case performs well in VHT_{queue}, but this comes at the expense of poorer density tracking, as shown by the episode-reward results in Figure 18.

Among the controlled strategies, PI-ALINEA optimised for $\bar{v}_{queue}$ achieves the lowest queue burden across the tested demand scenarios. By contrast, PI-ALINEA optimised for $\bar{v}$ achieves excellent mainline performance but can produce considerably worse queue-inclusive results, especially under higher demand noise. This highlights that marginal gains in mainline VHT are not necessarily meaningful when they are obtained by transferring delay to the on-ramp. The RL controller shows a similar trade-off: it generally produces one of the largest queue burdens among the metered controllers, approximately 32% higher than the best controlled alternative under clean demand, despite strong density-tracking performance. This is consistent with its reward formulation, which penalises deviations of ρ_3 from ρ_{des} but does not include a queue term. Exact numerical values are reported in Table 4.

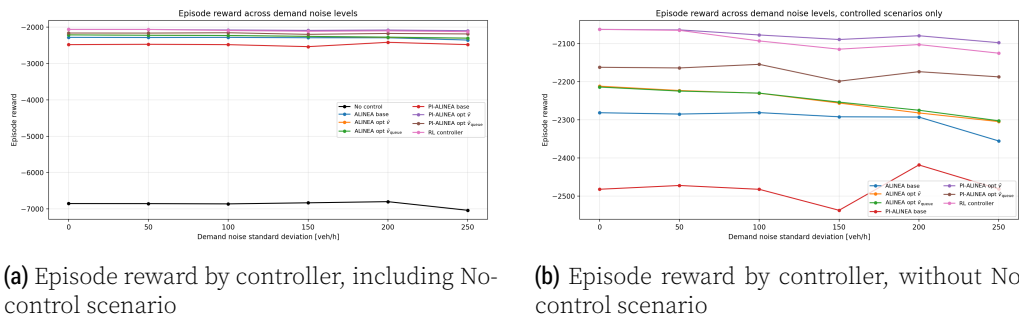


Figure 18. Evolution of episode reward by controller over increasing standard deviation of the demand

Figure 18 compares the episode reward across demand noise levels, both including and excluding the no-control scenario. Since the reward directly penalises deviations of the control-cell density from ρ_{des} , higher values indicate better tracking performance. The no-control scenario performs substantially worse than all controlled strategies, confirming that ramp metering is effective at regulating the control-cell density. Among the controlled scenarios, the RL controller and PI-ALINEA optimised for $\bar{v}$ achieve the best rewards under clean demand. Across increasing noise levels, both remain among the strongest performers, although PI-ALINEA optimised for $\bar{v}$ slightly outperforms the RL controller at higher noise levels. The original PI-ALINEA parameterisation performs

clearly worse than the optimised variants, confirming the benefit of retuning the benchmark controller. Exact indicator values can be found in Table 4.

Table 2. Percentage gap from best-performing controller per indicator and demand scenario. 0.0% indicates the best result; higher values indicate worse performance.

Controller	Noise std veh/h	VKT veh·km	VHT veh·h	VHT _{queue} veh·h	$\bar{v}$ km/h	$\bar{v}_{queue}$ km/h	Episode reward
No control	0	0.0%	34.1%	0.0%	25.4%	0.0%	232.2%
ALINEA ($K_R=10$)	0	0.0%	0.6%	34.3%	0.6%	25.5%	10.6%
ALINEA opt $\bar{v}$ ($K_R=23$)	0	0.0%	0.4%	23.3%	0.4%	18.9%	7.2%
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	0	0.0%	0.4%	23.0%	0.4%	18.7%	7.3%
PI-ALINEA ($K_R=100$, $K_P=4$)	0	0.0%	1.1%	35.4%	1.1%	26.1%	20.3%
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	0	0.0%	0.0%	30.6%	0.0%	23.4%	0.0%
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	0	0.0%	0.3%	13.3%	0.3%	11.8%	4.8%
RL controller (ep 263366)	0	0.0%	0.0%	50.2%	0.0%	33.4%	0.0%
No control	50	0.0%	33.8%	0.0%	25.2%	0.0%	232.1%
ALINEA ($K_R=10$)	50	0.0%	0.6%	34.3%	0.6%	25.6%	10.7%
ALINEA opt $\bar{v}$ ($K_R=23$)	50	0.0%	0.5%	24.9%	0.4%	19.9%	7.7%
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	50	0.0%	0.5%	24.8%	0.4%	19.8%	7.8%
PI-ALINEA ($K_R=100$, $K_P=4$)	50	0.0%	1.1%	35.5%	1.1%	26.2%	19.8%
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	50	0.0%	0.0%	36.7%	0.0%	26.8%	0.0%
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	50	0.0%	0.3%	13.9%	0.3%	12.2%	4.8%
RL controller (ep 263366)	50	0.0%	0.0%	53.3%	0.0%	34.8%	0.1%
No control	100	0.0%	33.9%	0.0%	25.3%	0.0%	230.3%
ALINEA ($K_R=10$)	100	0.0%	0.6%	35.7%	0.6%	26.3%	9.8%
ALINEA opt $\bar{v}$ ($K_R=23$)	100	0.0%	0.4%	28.5%	0.4%	22.2%	7.4%
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	100	0.0%	0.4%	27.3%	0.4%	21.5%	7.3%
PI-ALINEA ($K_R=100$, $K_P=4$)	100	0.0%	1.1%	36.2%	1.1%	26.6%	19.5%
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	100	0.0%	0.0%	35.5%	0.0%	26.2%	0.0%
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	100	0.0%	0.2%	15.1%	0.2%	13.1%	3.7%
RL controller (ep 263366)	100	0.0%	0.0%	49.0%	0.0%	32.9%	0.8%
No control	150	0.0%	32.9%	0.0%	24.8%	0.0%	227.0%
ALINEA ($K_R=10$)	150	0.0%	0.6%	33.1%	0.5%	24.9%	9.7%
ALINEA opt $\bar{v}$ ($K_R=23$)	150	0.0%	0.5%	29.5%	0.5%	22.8%	8.0%
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	150	0.0%	0.4%	27.9%	0.4%	21.8%	7.9%
PI-ALINEA ($K_R=100$, $K_P=4$)	150	0.0%	1.2%	34.7%	1.2%	25.7%	21.4%
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	150	0.0%	0.0%	33.7%	0.0%	25.2%	0.0%
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	150	0.0%	0.3%	18.1%	0.3%	15.3%	5.2%
RL controller (ep 263366)	150	0.0%	0.0%	41.9%	0.1%	29.6%	1.2%
No control	200	0.0%	31.8%	0.0%	24.1%	0.0%	227.0%
ALINEA ($K_R=10$)	200	0.0%	0.6%	34.8%	0.6%	25.8%	10.3%
ALINEA opt $\bar{v}$ ($K_R=23$)	200	0.0%	0.5%	39.6%	0.5%	28.4%	9.7%
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	200	0.0%	0.5%	38.8%	0.5%	28.0%	9.4%
PI-ALINEA ($K_R=100$, $K_P=4$)	200	0.0%	0.9%	33.5%	0.9%	25.1%	16.3%
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	200	0.0%	0.0%	49.8%	0.0%	33.2%	0.0%
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	200	0.0%	0.3%	23.9%	0.3%	19.3%	4.5%
RL controller (ep 263366)	200	0.0%	0.0%	43.3%	0.1%	30.2%	1.1%
No control	250	0.0%	35.0%	0.0%	25.9%	0.0%	235.5%
ALINEA ($K_R=10$)	250	0.0%	0.7%	39.0%	0.7%	28.0%	12.3%
ALINEA opt $\bar{v}$ ($K_R=23$)	250	0.0%	0.6%	32.5%	0.6%	24.5%	9.9%
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	250	0.0%	0.6%	30.9%	0.6%	23.6%	9.8%
PI-ALINEA ($K_R=100$, $K_P=4$)	250	0.0%	1.0%	36.6%	1.0%	26.8%	18.3%
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	250	0.0%	0.0%	48.1%	0.0%	32.5%	0.0%
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	250	0.0%	0.2%	28.3%	0.2%	22.0%	4.3%
RL controller (ep 263366)	250	0.0%	0.0%	44.4%	0.1%	30.8%	1.3%

4.5 Discussion

The reproduction confirms that the main simulation environment can be reconstructed with high fidelity. In particular, the no-control scenario and the reference PI-ALINEA scenario show close agreement with the figures reported in the original article. This suggests that the selected CTM discretisation, boundary conditions, demand profile, and lane-drop representation are sufficiently consistent with the original experimental setup.

However, the central performance claim of the original study is only partially supported. The reproduced RL controller achieves stable control-cell density tracking and smooth space–time density patterns, which is consistent with the qualitative behaviour reported by the authors. The original article describes the learned policy as maintaining the control target density close to the desired value with almost no fluctuations and showing some robustness under uncertain demands. Nevertheless, when compared against optimised benchmark controllers, the RL controller does not provide a clear advantage. In the reconstructed experiments, PI-ALINEA optimised for $\bar{v}$ matches or slightly outperforms the RL controller in mainline efficiency and episode reward, while PI-ALINEA optimised for $\bar{v}_{\text{queue}}$ performs substantially better when ramp queueing time is included. This result highlights the importance of benchmark calibration. The original study compares the proposed RL controller only against one PI-ALINEA parameterisation. In this reproduction, the same PI-ALINEA configuration was successfully reproduced, but the grid search showed that it is not the strongest available benchmark for the reconstructed environment. This makes the comparison with the RL controller sensitive to the choice of benchmark parameters. Once the benchmark controller is retuned, the RL policy appears less like a fundamentally superior control strategy and more like a slightly less queue-aware version of an optimised PI-ALINEA controller.

A second limitation concerns the reward design. The RL reward penalises only the deviation of the control-cell density ρ_3 from the desired density ρ_{des} . It therefore encourages density tracking at the downstream bottleneck, but does not directly penalise ramp queueing delay. This explains why the RL controller performs well in terms of episode reward and mainline VHT, but poorly in terms of $\text{VHT}_{\text{queue}}$. This is not necessarily a failure of the learning algorithm; rather, it reflects the objective that the agent was trained to optimise. The original authors themselves identify ramp queue management as a future extension, explicitly suggesting that queue length should be managed at the cost of some mainline efficiency.

The training setup also limits the interpretation of the learned policy. The ANN was trained on a single reconstructed demand profile, while robustness was evaluated only afterwards by adding demand noise. As a result, the trained policy may partly exploit regularities of this specific demand pattern rather than learning a broadly generalisable ramp-metering strategy. Training directly on randomised or noisy demand profiles would likely provide a more meaningful test of generalisation. In particular, demand perturbations could stimulate regions of the state space that are never visited under the deterministic base demand. For example, the ramp-demand encoding includes bins below 500 veh/h, although the deterministic ramp demand never falls below this value during training. Under sufficiently noisy demand, these bins may become active, allowing the agent to learn responses to states that are otherwise unused.

Finally, the reproduction was affected by several missing implementation details. The original article does not report the CTM discretisation parameters Δx and Δt , the demand profile numerically, the ϵ -decay schedule, the convergence criterion, the reward scaling constant, the network initialisation procedure, or the exact figure smoothing procedure. These omissions required several reconstruction choices and post-hoc checks. While the resulting baseline reproduction is visually close to the published figures, exact reproducibility remains limited. This supports the broader concern that transportation simulation studies require more complete reporting of source code, demand data, calibration procedures, and controller parameters.

5 Conclusion

This study reproduced the ramp-metering experiment of Zhou et al. using a reimplemented CTM simulation environment, benchmark controllers, and RL training pipeline. The no-control and reference PI-ALINEA scenarios were reproduced closely, indicating that the reconstructed simulation environment captures the main dynamics of the original study.

The reported superiority of the RL controller could not be confirmed once stronger benchmark controllers were introduced. The selected RL policy achieved stable density tracking and competitive mainline efficiency, but its performance was comparable to PI-ALINEA optimised for $\bar{v}$ and clearly inferior to PI-ALINEA optimised for $\bar{v}_{\text{queue}}$ when ramp queueing delay was included. The RL controller therefore appears effective for the objective it was trained on, namely control-cell density tracking, but not necessarily superior from a system-wide perspective.

The results emphasise three main lessons. First, benchmark controllers must be carefully calibrated before claims of RL superiority can be assessed. Second, reward design strongly determines the behaviour of learned controllers; omitting queue delay encourages policies that may stabilise the mainline by transferring delay to the ramp. Third, reproducibility requires more complete reporting than was provided in the original article, particularly regarding demand profiles, discretisation choices, learning schedules, and checkpoint-selection procedures. Most importantly, quantitative performance indicators should be reported alongside graphical results to enable direct comparison.

Future work should train the RL agent on randomised demand profiles, include ramp queueing delay directly in the reward function, and compare against optimised classical controllers under multiple demand scenarios and random seeds. Such an evaluation would better determine whether value-function approximation provides a genuine advantage over well-tuned feedback ramp-metering controllers for distant downstream bottlenecks.

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A Software and Computational Environment

All experiments were conducted in Python. The key software dependencies used in this replication are listed in Table 3. The complete environment specification, including all transitive dependencies, is provided in the accompanying code repository.

Table 3. Software dependencies used in this replication.

Package	Version	Purpose
Python	3.10.10	Programming language
NumPy	2.2.6	Numerical computation
PyTorch	2.12.0	Neural network implementation
Matplotlib	3.10.9	Visualisation
SciPy	1.15.3	Signal processing and interpolation
tqdm	4.67.3	Training progress tracking

Hardware. Experiments were run on a machine equipped with an Intel(R) Core(TM) Ultra 7 265 (20 cores) processor (dedicated CPU usage oscillated between 95% and 100%), 32 GB of RAM, and Intel(R) Graphics graphics card. Training of the RL agent required approximately 55 hours for 700'000 episodes.

B Controller performances

B.1 Computation of Percentage Gaps

To compare controllers across indicators with different units and magnitudes, relative percentage gaps were computed with respect to the best-performing controller for each demand scenario and performance indicator. The convention was chosen such that 0.0% always denotes the best observed result, while positive values indicate worse performance relative to that best result.

For indicators where higher values are better, the percentage gap was computed as:

$$\text{Gap} = \frac{x_{\text{best}} - x}{|x_{\text{best}}|} \cdot 100 \quad (20)$$

where x is the value obtained by a given controller and x_{best} is the highest value observed among all controllers for the same demand scenario.

For indicators where lower values are better, the percentage gap was computed as:

$$\text{Gap} = \frac{x - x_{\text{best}}}{|x_{\text{best}}|} \cdot 100 \quad (21)$$

where x_{best} is the lowest value observed among all controllers for the same demand scenario.

The higher-is-better indicators are VKT, $\bar{v}$, $\bar{v}_{\text{queue}}$, and episode reward. The lower-is-better indicators are VHT and $\text{VHT}_{\text{queue}}$.

For the episode reward, larger values are better because the reward is defined as a negative penalty on deviations from the desired control-cell density. Thus, a value closer to zero represents better density tracking. Results are available in Table 2.

B.2 Full controller performance results for each indicator and noise demand scenario

Table 4. Full controller performance results across base and noisy demand scenarios.

Controller	Noise std veh/h	VKT veh·km	VHT veh·h	VHT _{queue} veh·h	$\bar{v}$ km/h	$\bar{v}_{queue}$ km/h	Episode reward
No control	0	27700.62	309.79	309.79	89.42	89.42	-6852.17
ALINEA ($K_R=10$)	0	27700.62	232.49	415.94	119.15	66.60	-2281.23
ALINEA opt $\bar{v}$ ($K_R=23$)	0	27700.62	232.05	381.98	119.37	72.52	-2211.65
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	0	27700.62	232.07	381.00	119.36	72.70	-2213.92
PI-ALINEA ($K_R=100$, $K_P=4$)	0	27700.62	233.74	419.41	118.51	66.05	-2481.82
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	0	27700.62	231.12	404.59	119.85	68.47	-2062.83
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	0	27700.62	231.74	351.09	119.53	78.90	-2162.09
RL controller (ep 263366)	0	27696.70	231.09	465.37	119.85	59.52	-2062.83
No control	50	27690.87	309.06	309.06	89.60	89.60	-6854.04
ALINEA ($K_R=10$)	50	27690.87	232.42	415.22	119.14	66.69	-2284.73
ALINEA opt $\bar{v}$ ($K_R=23$)	50	27690.87	232.03	385.96	119.34	71.75	-2222.98
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	50	27690.87	232.04	385.56	119.34	71.82	-2224.51
PI-ALINEA ($K_R=100$, $K_P=4$)	50	27690.87	233.59	418.68	118.55	66.14	-2471.97
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	50	27690.87	231.04	422.34	119.85	65.57	-2064.03
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	50	27690.87	231.66	352.03	119.53	78.66	-2163.82
RL controller (ep 263366)	50	27684.80	230.98	473.81	119.86	58.43	-2065.24
No control	100	27619.04	308.53	308.53	89.52	89.52	-6861.00
ALINEA ($K_R=10$)	100	27619.04	231.74	418.67	119.18	65.97	-2280.96
ALINEA opt $\bar{v}$ ($K_R=23$)	100	27619.04	231.42	396.43	119.34	69.67	-2230.15
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	100	27619.04	231.42	392.87	119.35	70.30	-2229.82
PI-ALINEA ($K_R=100$, $K_P=4$)	100	27619.04	233.00	420.11	118.54	65.74	-2481.98
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	100	27619.04	230.47	418.16	119.84	66.05	-2077.39
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	100	27619.04	230.95	355.10	119.59	77.78	-2154.37
RL controller (ep 263366)	100	27612.90	230.51	459.77	119.79	60.06	-2093.03
No control	150	27573.66	305.89	305.89	90.14	90.14	-6831.61
ALINEA ($K_R=10$)	150	27573.66	231.39	407.10	119.17	67.73	-2291.89
ALINEA opt $\bar{v}$ ($K_R=23$)	150	27573.66	231.16	395.99	119.28	69.63	-2255.83
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	150	27573.66	231.15	391.24	119.29	70.48	-2253.44
PI-ALINEA ($K_R=100$, $K_P=4$)	150	27573.66	232.92	411.97	118.38	66.93	-2536.98
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	150	27573.66	230.12	408.89	119.82	67.44	-2089.26
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	150	27573.66	230.80	361.12	119.47	76.36	-2198.59
RL controller (ep 263366)	150	27564.70	230.20	434.14	119.74	63.49	-2114.65
No control	200	27673.15	304.38	304.38	90.92	90.92	-6799.73
ALINEA ($K_R=10$)	200	27673.15	232.30	410.42	119.13	67.43	-2292.58
ALINEA opt $\bar{v}$ ($K_R=23$)	200	27673.15	232.23	425.02	119.16	65.11	-2281.76
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	200	27673.15	232.19	422.52	119.18	65.49	-2274.67
PI-ALINEA ($K_R=100$, $K_P=4$)	200	27673.15	233.08	406.37	118.73	68.10	-2418.12
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	200	27673.15	230.97	455.91	119.81	60.70	-2079.41
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	200	27673.15	231.56	377.14	119.51	73.38	-2173.60
RL controller (ep 263366)	200	27668.50	231.08	436.15	119.74	63.44	-2102.57
No control	250	27630.56	311.39	311.39	88.73	88.73	-7037.18
ALINEA ($K_R=10$)	250	27630.56	232.32	432.74	118.94	63.85	-2355.20
ALINEA opt $\bar{v}$ ($K_R=23$)	250	27630.56	232.00	412.64	119.10	66.96	-2304.48
ALINEA opt $\bar{v}_{queue}$ ($K_R=21$)	250	27630.56	231.99	407.66	119.10	67.78	-2302.28
PI-ALINEA ($K_R=100$, $K_P=4$)	250	27630.56	233.10	425.50	118.53	64.94	-2481.05
PI-ALINEA opt $\bar{v}$ ($K_R=2$, $K_P=93$)	250	27630.56	230.71	461.15	119.76	59.92	-2097.64
PI-ALINEA opt $\bar{v}_{queue}$ ($K_R=2$, $K_P=56$)	250	27630.56	231.27	399.37	119.48	69.19	-2187.24
RL controller (ep 263366)	250	27622.60	230.80	449.73	119.68	61.42	-2125.13